

Byzantine Robustness Comparison: CNN vs. SNN (Simplest Sweeps)

ByzFL SNN Benchmark

June 25, 2026

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1 Introduction

This document presents side-by-side comparison grids of Byzantine robustness results for CNN and SNN architectures under the simplest sweep configurations. The configurations compared are:

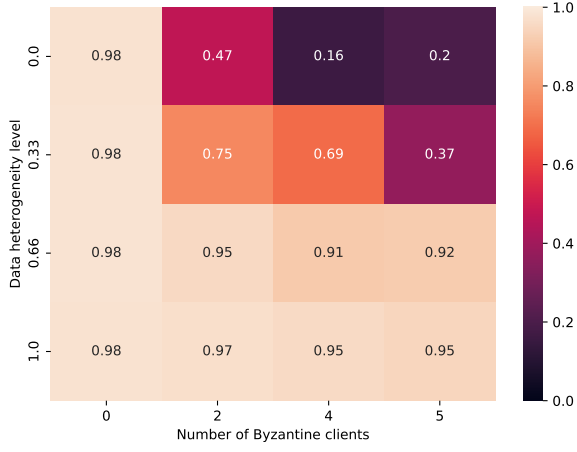
1. **CNN**: Convolutional Neural Network on MNIST ('cnn-simplest-plots')
2. **SNN**: Spiking Neural Network on MNIST ('snn-simplest-plots')

Each grid illustrates the test accuracy under varying numbers of Byzantine clients ($f \in [0, 2, 4, 5]$) and data heterogeneity levels ($\gamma \in [0.0, 0.33, 0.66, 1.0]$) with $N = 10$ honest clients.

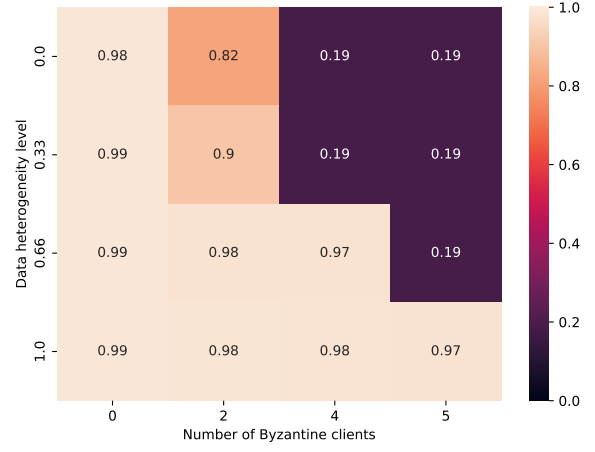
2 Worst-Case Performance (Across All Attacks)

This section compares the test accuracy under the worst-case attack scenario (minimizing the maximum damage over all attacks) for both the dynamically selected best aggregator and each of the fixed aggregators.

2.1 Best Aggregator Choice



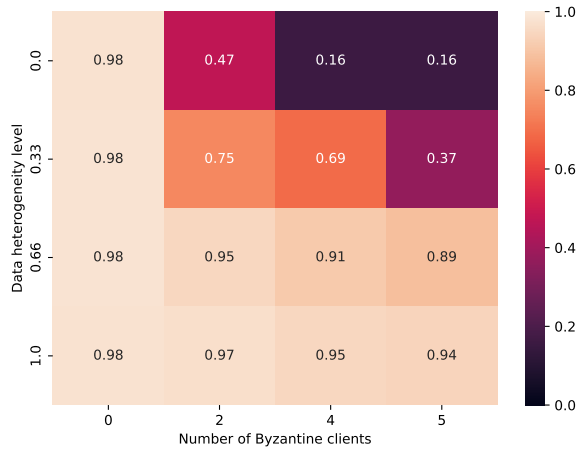
(a) CNN (Best Aggregator Choice)



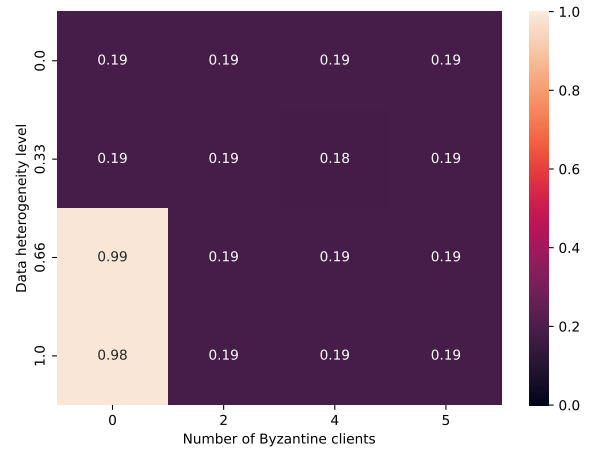
(b) SNN (Best Aggregator Choice)

Figure 1: Worst-case test accuracy comparison using the best overall aggregator selection.

2.2 Centered Clipping



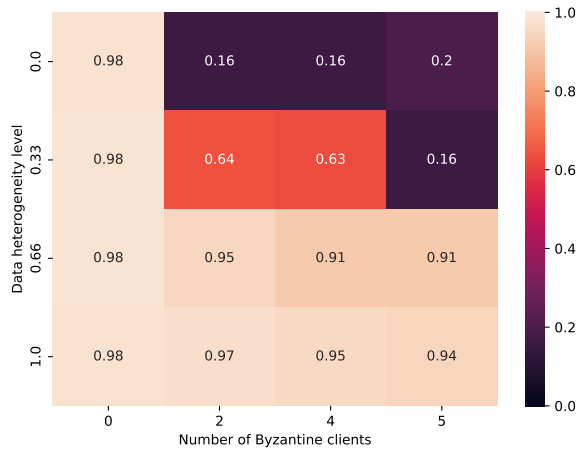
(a) CNN (Centered Clipping)



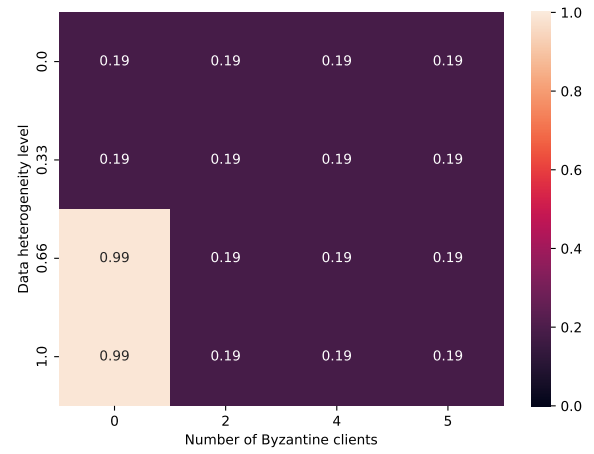
(b) SNN (Centered Clipping)

Figure 2: Worst-case test accuracy comparison using Centered Clipping.

2.3 Geometric Median



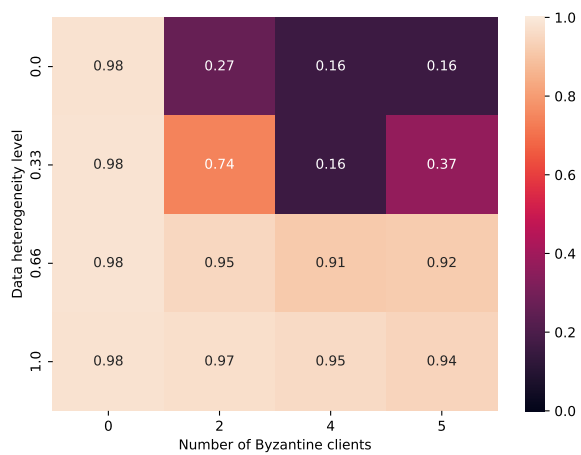
(a) CNN (Geometric Median)



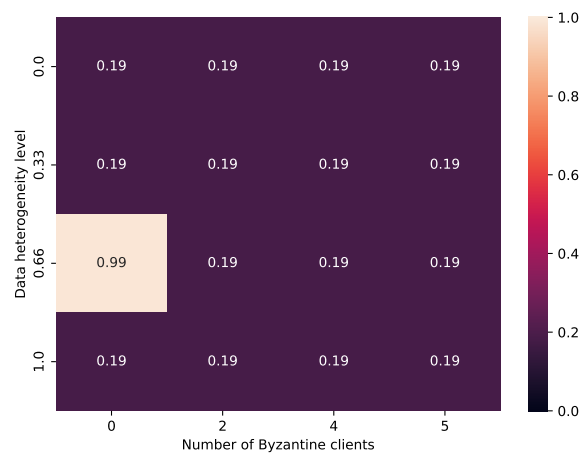
(b) SNN (Geometric Median)

Figure 3: Worst-case test accuracy comparison using Geometric Median.

2.4 Multi-Krum



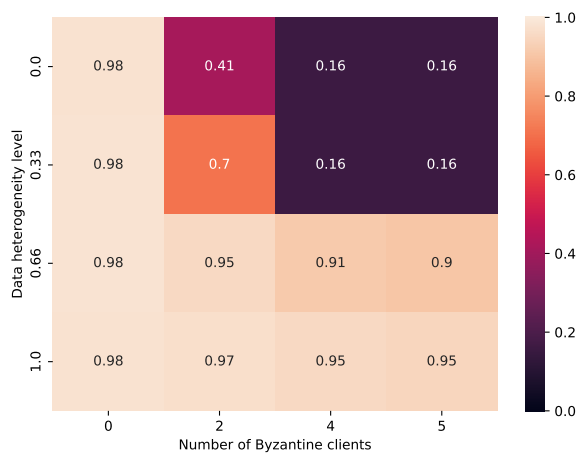
(a) CNN (Multi-Krum)



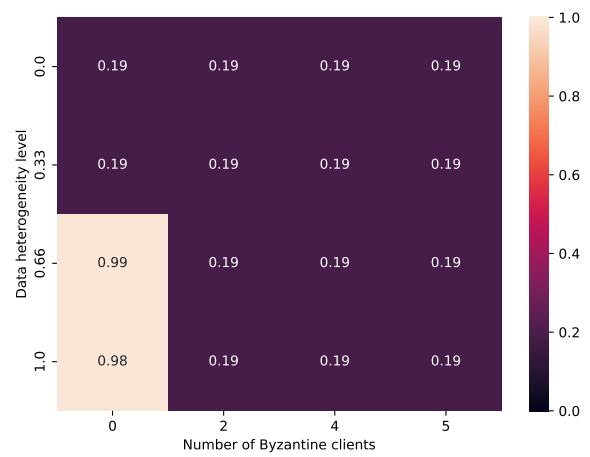
(b) SNN (Multi-Krum)

Figure 4: Worst-case test accuracy comparison using Multi-Krum.

2.5 Trimmed Mean



(a) CNN (Trimmed Mean)



(b) SNN (Trimmed Mean)

Figure 5: Worst-case test accuracy comparison using Trimmed Mean.

3 Fixed Attack: Optimal A Little Is Enough (ALIE)

This section compares performance under the Optimal ALIE Byzantine attack, showing both the dynamically selected best aggregator and each of the fixed aggregators.

3.1 Best Aggregator Choice

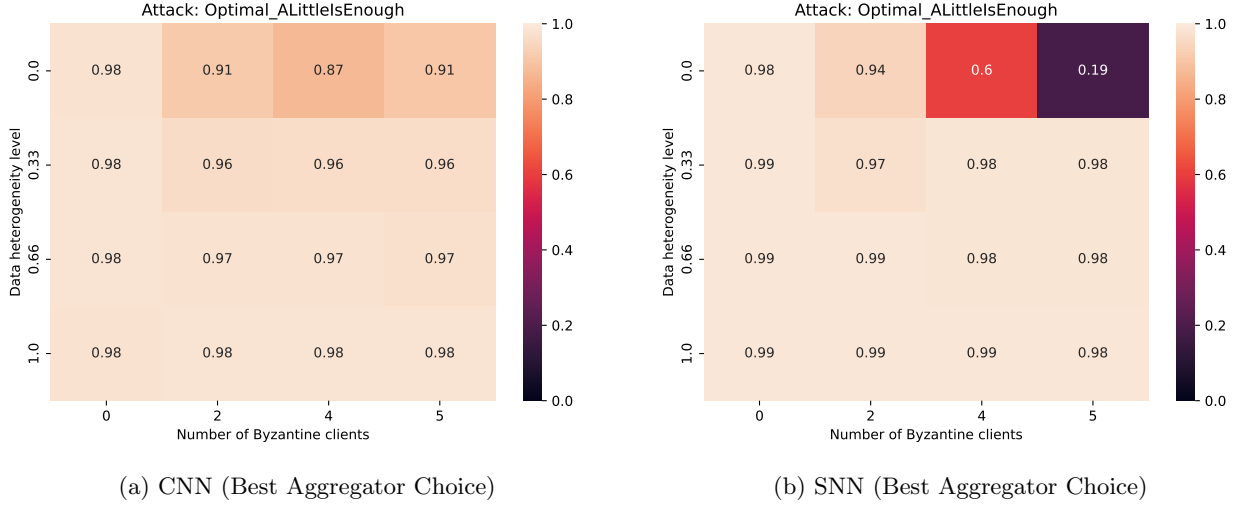
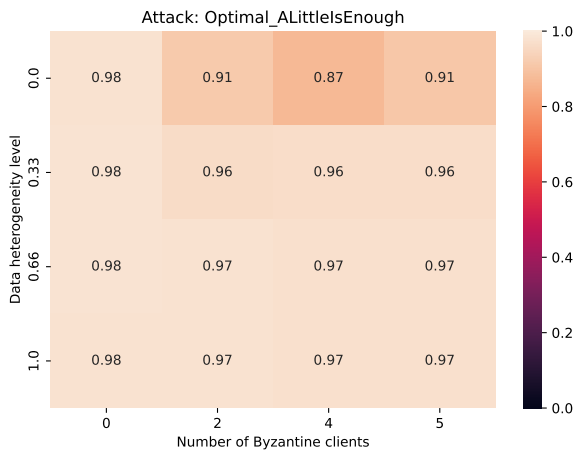
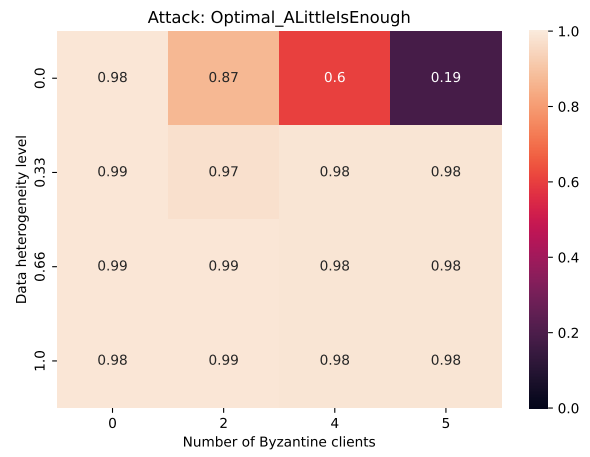


Figure 6: Accuracy comparison under Optimal ALIE attack using the best aggregator selection.

3.2 Centered Clipping



(a) CNN (Centered Clipping)



(b) SNN (Centered Clipping)

Figure 7: Accuracy comparison under Optimal ALIE attack using Centered Clipping.

3.3 Geometric Median

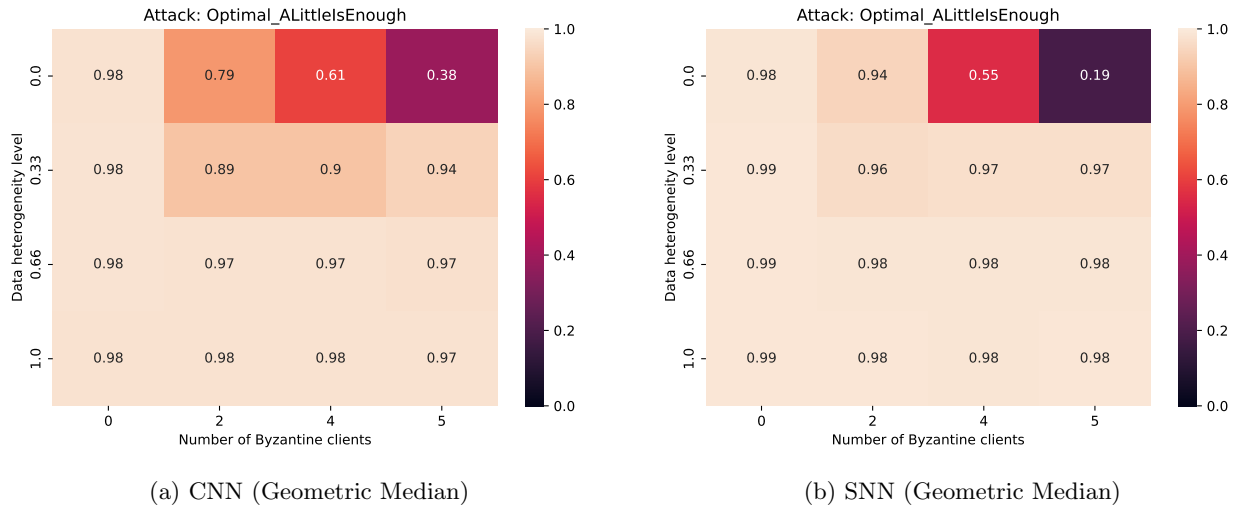
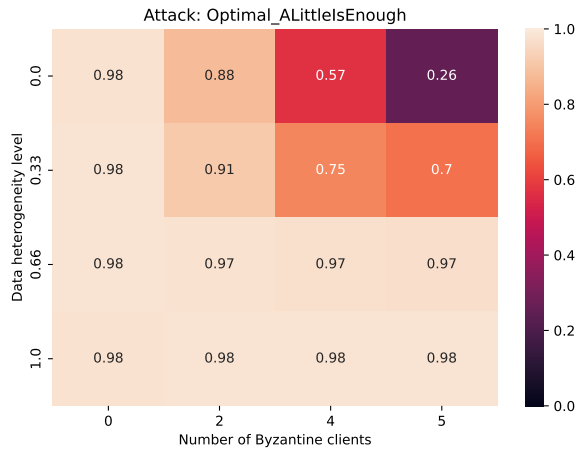
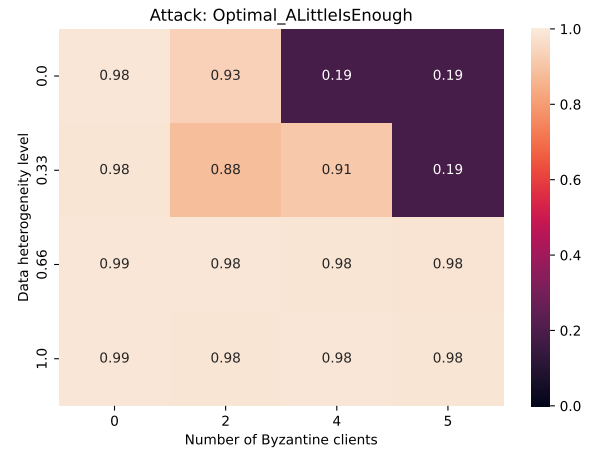


Figure 8: Accuracy comparison under Optimal ALIE attack using Geometric Median.

3.4 Multi-Krum



(a) CNN (Multi-Krum)



(b) SNN (Multi-Krum)

Figure 9: Accuracy comparison under Optimal ALIE attack using Multi-Krum.

3.5 Trimmed Mean

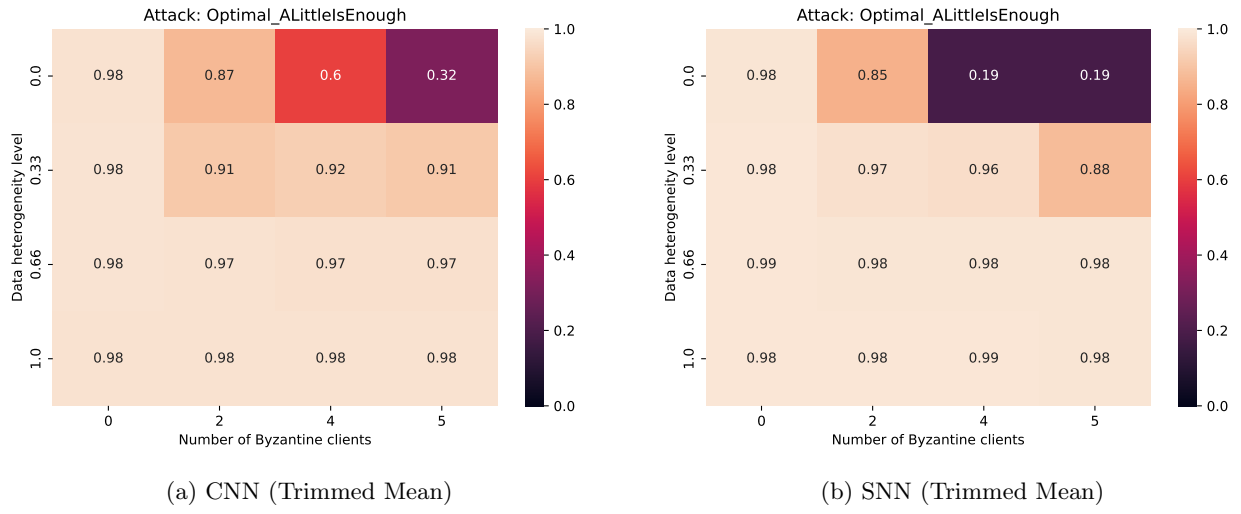


Figure 10: Accuracy comparison under Optimal ALIE attack using Trimmed Mean.

4 Fixed Attack: Optimal Inner Product Manipulation (IPM)

This section compares performance under the Optimal IPM Byzantine attack, showing both the dynamically selected best aggregator and each of the fixed aggregators.

4.1 Best Aggregator Choice

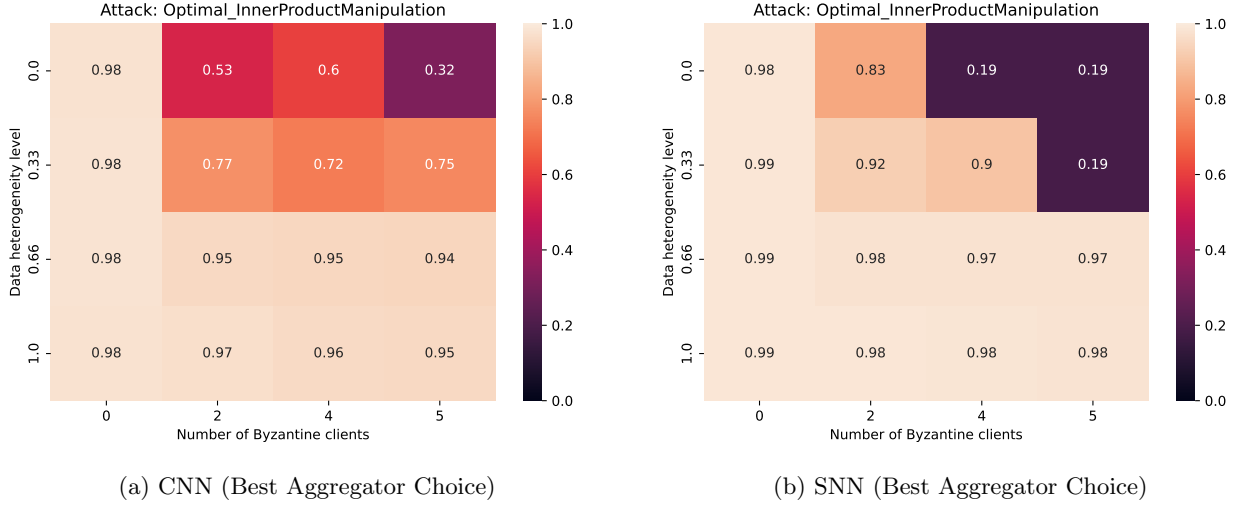


Figure 11: Accuracy comparison under Optimal IPM attack using the best aggregator selection.

4.2 Centered Clipping

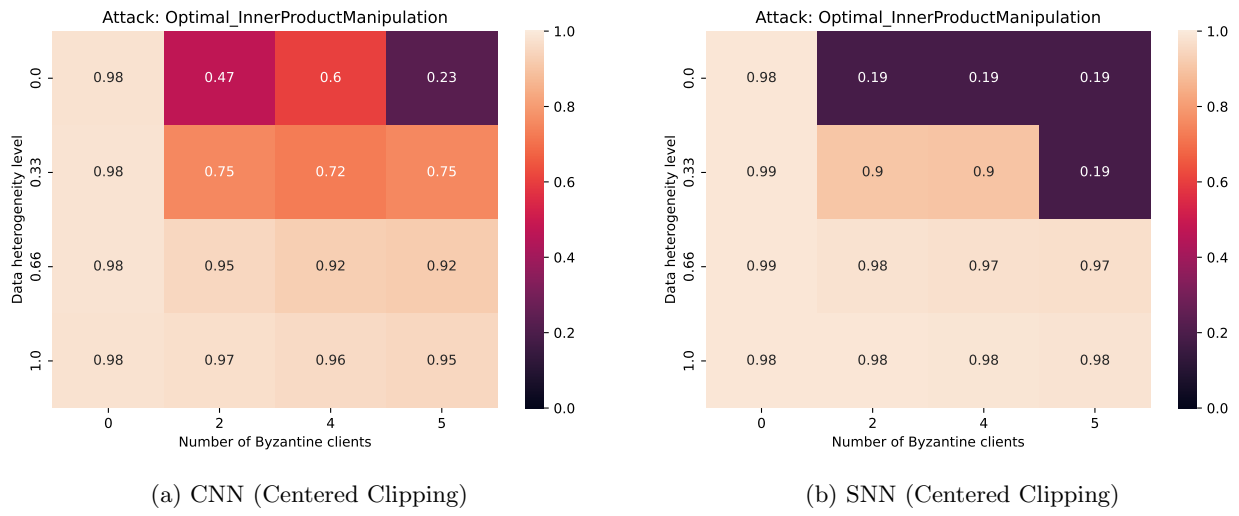
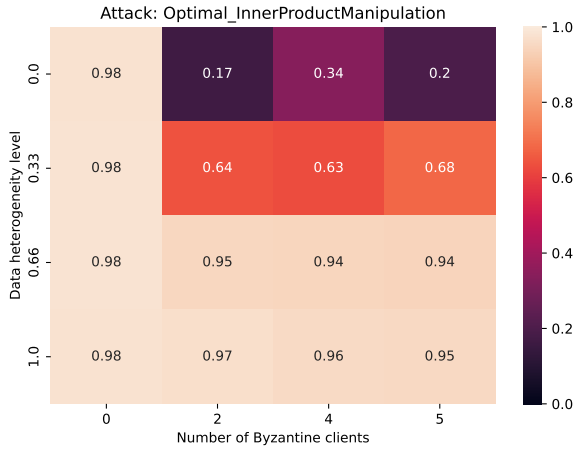
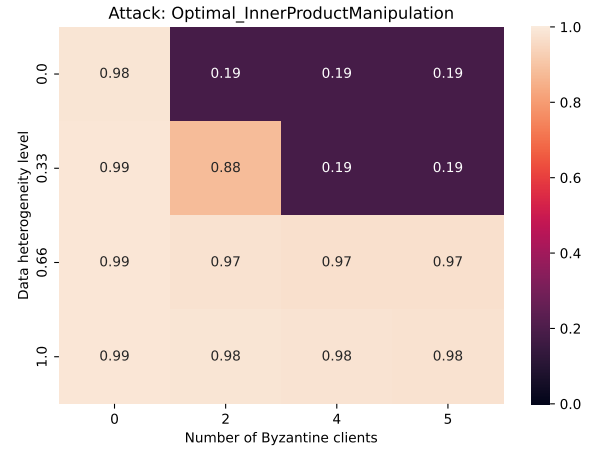


Figure 12: Accuracy comparison under Optimal IPM attack using Centered Clipping.

4.3 Geometric Median



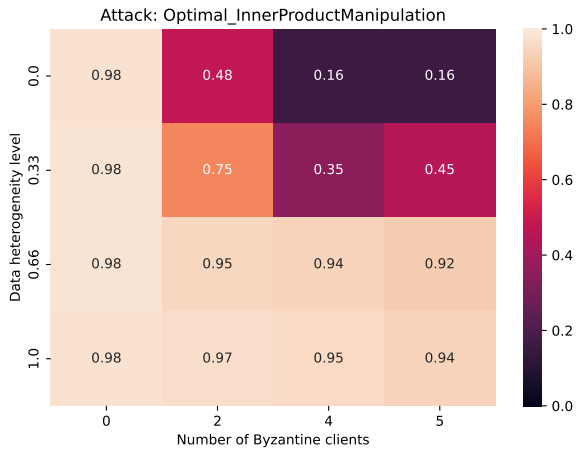
(a) CNN (Geometric Median)



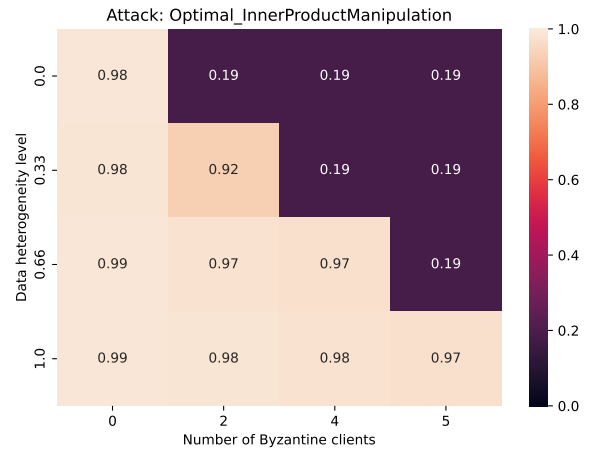
(b) SNN (Geometric Median)

Figure 13: Accuracy comparison under Optimal IPM attack using Geometric Median.

4.4 Multi-Krum



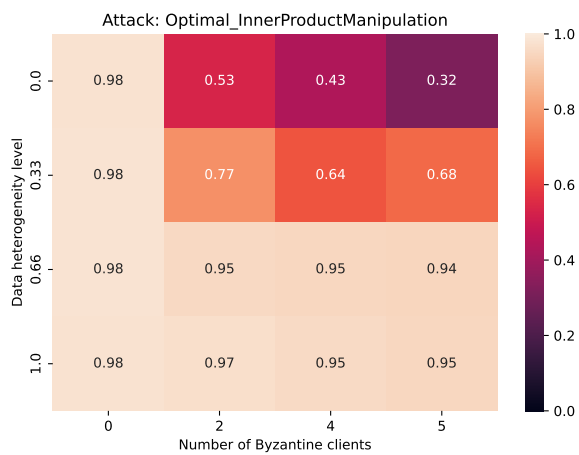
(a) CNN (Multi-Krum)



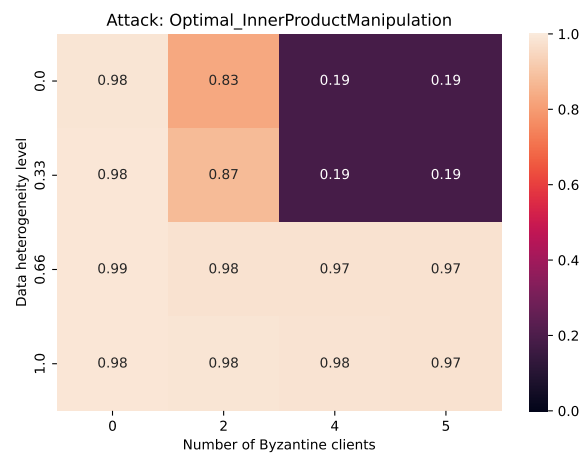
(b) SNN (Multi-Krum)

Figure 14: Accuracy comparison under Optimal IPM attack using Multi-Krum.

4.5 Trimmed Mean



(a) CNN (Trimmed Mean)



(b) SNN (Trimmed Mean)

Figure 15: Accuracy comparison under Optimal IPM attack using Trimmed Mean.

5 Fixed Attack: Sign Flipping

This section compares performance under the Sign Flipping Byzantine attack, showing both the dynamically selected best aggregator and each of the fixed aggregators.

5.1 Best Aggregator Choice



(a) CNN (Best Aggregator Choice)



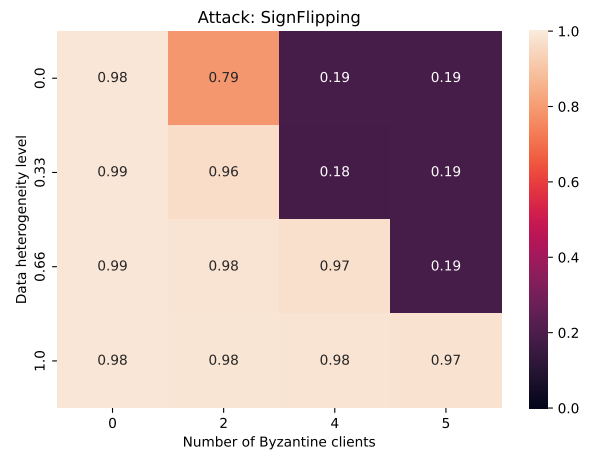
(b) SNN (Best Aggregator Choice)

Figure 16: Accuracy comparison under Sign Flipping attack using the best aggregator selection.

5.2 Centered Clipping



(a) CNN (Centered Clipping)



(b) SNN (Centered Clipping)

Figure 17: Accuracy comparison under Sign Flipping attack using Centered Clipping.

5.3 Geometric Median

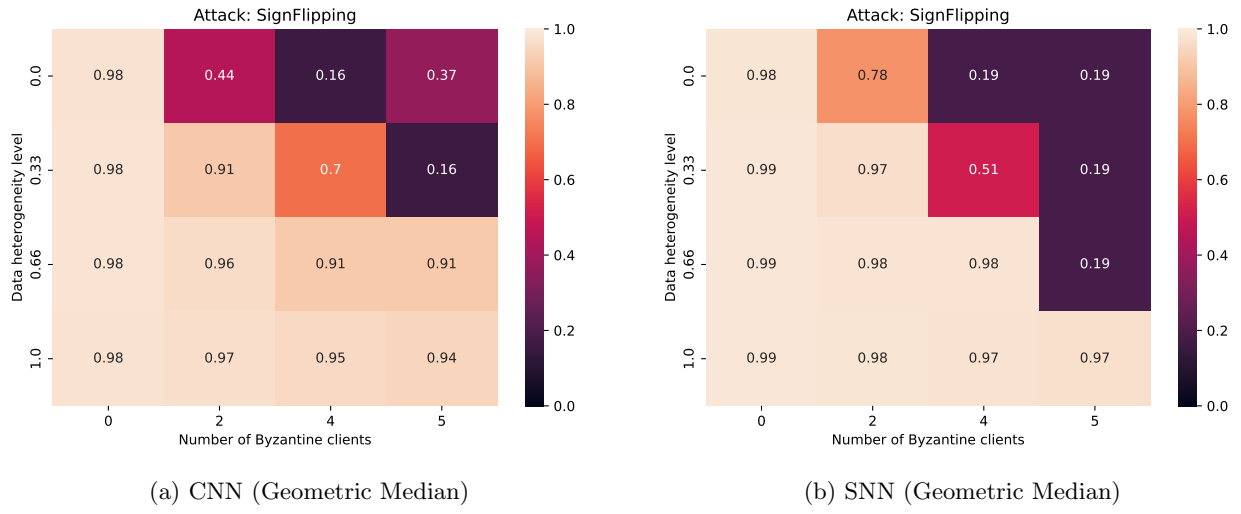
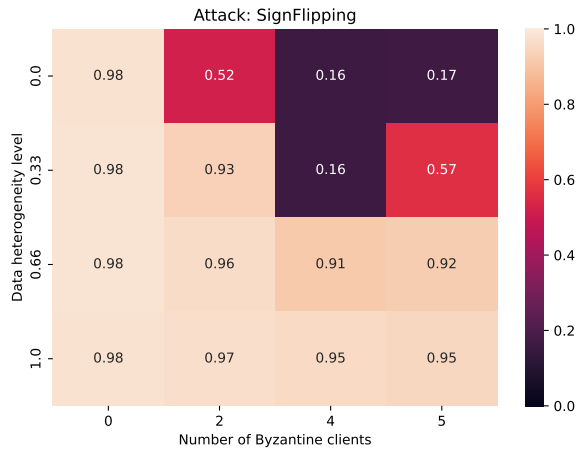


Figure 18: Accuracy comparison under Sign Flipping attack using Geometric Median.

5.4 Multi-Krum



(a) CNN (Multi-Krum)



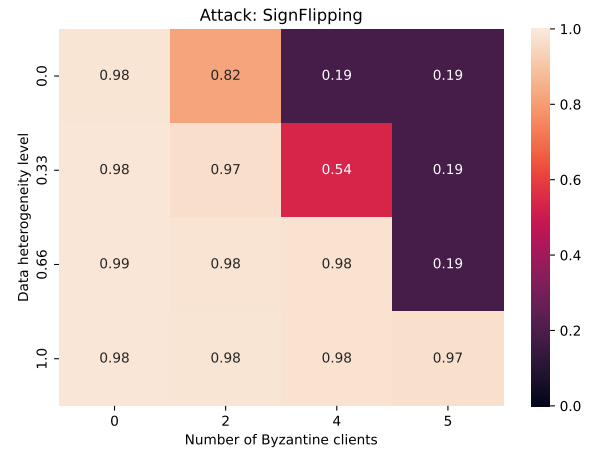
(b) SNN (Multi-Krum)

Figure 19: Accuracy comparison under Sign Flipping attack using Multi-Krum.

5.5 Trimmed Mean



(a) CNN (Trimmed Mean)



(b) SNN (Trimmed Mean)

Figure 20: Accuracy comparison under Sign Flipping attack using Trimmed Mean.